

Thermoelectric Refrigerator



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As summers are approaching, everyone would need a refrigerator. In many cases, people cannot afford a regular-size refrigerator. Especially students staying in a hostel can neither afford it nor would like to carry such a big appliance along with them when they leave. Also, these

refrigerators consume power to the tune of 500W, which is not allowed in most hostels.

Presented here is a thermoelectric refrigerator that can be built with easily available off-the-shelf components for approximately ₹ 1200. The refrigerator consumes only around 60W of power. There is also a control system in the refrigerator that monitors and

controls the temperature. The overall size is small, so it can be kept quite comfortably in a small hostel room. Fig. 1 shows the author's prototype and the rough overall architecture.

Circuit and working

Fig. 2 shows the circuit diagram of the thermoelectric refrigerator. The

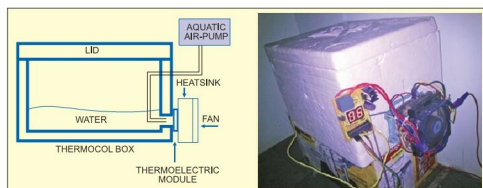


Fig. 1: Screenshot of the author's prototype and the rough overall architecture

Test Points

Test point	Details
TP0	0V, GND
TP1	5V (with regards to TP0)
TP2	12V (with regards to TP0)
TP3	Voltage corresponding to temperature (10mV/°C)
TP4	Pulse when the program is running
TP5	Pulse when the program is running
TP6	High to enable the relay RL1
TP7	Low when S1 is pressed

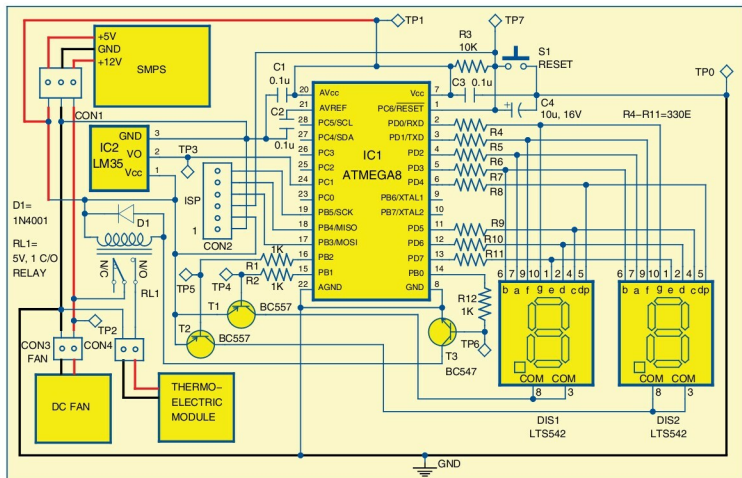


Fig. 2: Circuit diagram of the thermoelectric refrigerator